

INGA, Russian Federation

WMO: 307030

Lat: 52.971N

Lon: 101.959E

Elev: 1782

StdP: 13.77

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-38.4	-33.4	-42.6	0.5	-38.0	-37.8	0.7	-33.1	17.5	10.1	12.8	8.3	0.2	230	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.3	82.2	64.8	79.3	63.6	76.5	62.0	67.7	77.1	66.2	75.9	64.6	73.6	5.0	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.6	97.5	72.3	62.8	91.3	70.4	61.0	85.8	68.8	33.3	76.9	32.0	76.1	30.7	74.1	73.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.3	11.2	9.1	-46.1	86.9	4.8	2.8	-49.5	89.0	-52.3	90.6	-54.9	92.2	-58.4	94.2		
(5)			-46.1	70.5	4.7	1.4	-49.5	71.5	-52.2	72.3	-54.9	73.0	-58.3	74.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	29.8	-7.2	-0.9	16.6	35.1	45.9	58.6	62.8	58.9	46.1	31.7	11.4	-3.5
	DBStd	26.38	10.84	12.31	11.85	8.90	7.15	6.47	4.32	5.75	7.15	9.14	12.74	13.05
	HDD50	8412	1773	1425	1034	452	169	8	0	7	160	568	1158	1658
	HDD65	12901	2238	1845	1499	898	592	207	93	199	566	1033	1608	2123
	CDD50	1038	0	0	0	3	43	267	397	284	44	0	0	0
	CDD65	51	0	0	0	0	0	16	25	10	0	0	0	0
	CDH74	1151	0	0	0	3	106	446	375	202	19	0	0	0
	CDH80	192	0	0	0	0	19	92	50	29	2	0	0	0
	WSAvg	3.0	1.6	2.3	3.1	4.0	4.8	3.7	3.4	3.1	3.2	2.8	2.2	1.7
(15) Precipitation	PrecAvg	20.6	0.6	0.5	0.7	1.1	1.6	3.3	4.0	3.7	2.0	1.1	1.1	0.9
	PrecMax	31.0	1.3	0.9	2.2	2.4	3.5	5.7	12.0	7.8	4.2	1.8	2.2	1.8
	PrecMin	13.5	0.1	0.1	0.3	0.5	0.4	1.5	1.4	1.3	0.7	0.4	0.4	0.3
	PrecStd	3.9	0.3	0.2	0.4	0.4	0.6	1.3	2.1	1.7	0.9	0.3	0.4	0.4
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	24.9	34.3	51.6	72.0	83.8	86.7	85.1	83.2	76.5	64.0	43.6	29.5
		MCWB	22.7	27.5	39.8	49.5	58.2	65.0	66.6	67.1	57.7	47.2	35.8	27.4
	2%	DB	19.8	28.2	45.0	63.9	77.1	82.5	81.0	79.2	70.9	57.2	36.7	23.4
		MCWB	18.1	23.9	35.0	46.8	55.0	64.1	66.3	65.5	56.9	44.7	32.5	21.8
	5%	DB	15.2	23.7	40.5	58.2	72.0	79.2	78.3	75.6	66.3	51.5	32.7	19.7
		MCWB	13.8	20.6	32.6	43.3	51.9	62.8	65.5	63.9	54.0	41.7	30.2	18.6
	10%	DB	10.8	19.2	35.6	52.5	66.3	75.9	75.5	72.2	60.8	46.1	29.6	15.7
		MCWB	9.7	16.7	29.8	40.7	49.0	60.8	64.1	62.4	51.8	39.6	27.4	14.7
	0.4%	WB	22.9	28.0	39.5	51.5	60.3	68.3	70.7	68.9	62.3	49.0	36.4	27.4
		MCDB	24.4	32.6	50.1	67.3	80.8	80.5	78.5	78.4	72.8	62.4	41.5	29.9
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	18.0	24.2	36.0	47.7	56.0	66.1	68.6	66.9	58.3	45.4	33.0	21.9
		MCDB	19.8	27.8	44.3	61.8	73.4	78.1	76.9	76.7	66.9	55.7	36.1	23.2
	5%	WB	13.7	20.8	33.1	44.0	52.8	64.4	66.9	65.1	55.3	42.6	30.5	18.6
		MCDB	15.0	23.7	39.3	56.2	69.5	76.8	75.3	74.0	64.4	50.4	32.8	19.7
	10%	WB	9.7	16.7	30.2	41.0	50.1	62.4	65.2	63.0	52.2	39.6	27.4	14.7
		MCDB	10.8	19.1	35.3	51.3	64.2	73.6	73.4	70.6	60.0	45.1	29.5	15.6
	5% DB	MDBR	25.6	31.3	30.4	25.0	27.9	25.8	21.3	21.3	24.2	23.3	22.6	22.1
		MCDBR	26.4	33.3	32.5	37.0	42.3	33.8	28.7	29.5	33.8	32.7	22.8	21.7
		MCWBR	24.7	28.6	22.8	20.1	21.2	17.2	15.5	16.4	19.1	20.6	18.3	20.3
		MCWBR	26.1	32.0	28.6	34.4	38.6	29.9	23.9	26.5	30.3	30.9	22.0	21.7
(35) Mean Daily Temperature Range	5% WB	MCWBR	24.4	27.6	20.2	19.7	20.3	16.2	14.0	15.4	18.0	20.0	18.2	20.3
	taub		0.270	0.300	0.341	0.386	0.431	0.423	0.419	0.389	0.343	0.321	0.297	0.272
		taud	2.125	2.107	2.122	2.193	2.105	2.245	2.295	2.361	2.440	2.390	2.211	2.092
(42) Clear-Sky Solar Irradiance	Ebn at Noon		220	250	264	267	259	263	262	264	265	245	208	191
		Edh at Noon	22	31	38	41	47	41	39	34	28	24	20	19
	RadAvg		364	692	1162	1535	1717	1820	1661	1444	1071	674	376	263
(45) All-Sky Solar Radiation	RadStd		22	40	58	72	106	104	144	89	89	43	28	19

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (10 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page